

Pavithra Priyadarshini Selvakumar

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Academic Profile

Environmental policy and management scholar examining how institutions, policy actors, communities, and emerging technologies shape environmental decision-making, policy learning, implementation, participation, sustainability, resilience, and equitable responses to environmental change. Research spans U.S. settings and an emerging international research agenda and integrates policy-process theory, community-engaged research, and qualitative and computational methods.

Academic Appointments**Postdoctoral Research Scientist**

Climate School, Columbia University

September 2025 – Present

New York City, New York

Education**Doctor of Philosophy in Environmental Science**

Oklahoma State University

June 2025

Stillwater, Oklahoma

Dissertation: Institutional Sustainability in U.S. Land-Grant Universities:

A Comparative Study of Oklahoma State University (OSU) and

Colorado State University (CSU)

GPA: 4.0/4.0

Master of Science in Environmental Assessment and Management

University of East Anglia

September 2018

Norwich, United Kingdom

Bachelor of Technology in Information Technology

Anna University

May 2015

Chennai, India

Research Interests

Environmental Policy and Management; Environmental Policy and Governance; Policy Process and Policy Learning; Climate and Energy Policy; Environmental Sustainability and Resilience; Collaborative and Community Governance; Environmental Justice; AI and Emerging Technology Governance

Research Experience**Resilient Coastal Communities Project**

- Lead research within the Resilient Coastal Communities Project in collaboration with the New York City Environmental Justice Alliance and the Center for Sustainable Urban Development.
- Study how trust, continuity, shared authority, and reciprocal accountability support meaningful and durable community engagement in coastal resilience planning.
- Mentor student research on community-centered flood resilience and East Harlem organizations' expectations of U.S. Army Corps of Engineers planning processes.

Urban Flood-Risk Communication in Douala, Cameroon

- Serve as Principal Investigator for a \$49,500 Columbia Climate School seed-funded international, community-engaged public policy project, A Knowledge, Attitudes, and Practices-Informed Flood Risk Communication Pilot in Douala, Cameroon.

- Examine how flood-risk knowledge, institutional trust, preparedness practices, and communication systems shape urban flood-risk governance and community preparedness.
- Develop research design, IRB protocols, the KAP survey, local project coordination, and the development of context-specific flood-risk communication strategies.

AI, Climate Justice, and Environmental Governance

- Conduct research through the Laboratory for AI, Climate and Society on the responsible use of AI in climate and environmental decision-making.
- Develop a feminist research agenda examining the intersections of AI, climate justice, gender, and environmental governance.
- Collaborate with interdisciplinary researchers to develop publications, research initiatives, and grant proposals addressing the societal implications of AI.

Doctoral Dissertation Research — Institutional Sustainability Oklahoma State University

August 2021 – June 2025
Stillwater, Oklahoma

- Conducted semi-structured qualitative interviews with diverse stakeholders from OSU and CSU, including staff, faculty, students, and community members, to collect data on campus sustainability practices.
- Performed iterative coding based on drivers, barriers, leadership, and other critical factors influencing sustainable practices in higher education institutions.
- Conducted Group Model Building (GMB) sessions with stakeholders to gain insights into their perceptions, challenges, and suggestions regarding sustainability efforts on campus.
- Analyzed data using software such as NVivo to identify patterns, trends, and areas for improvement in campus sustainability efforts.
- Developed recommendations for enhancing sustainability practices on campus based on research findings.

Graduate Research Assistant – NSF EPSCoR Social Dynamics research team

August 2021 – May 2025
Oklahoma

- Worked with Opinion Leader Advisory Network (OLAN) and Extended Peer Scientific Advisory Network (EPSAN) to provide socially sustainable solutions to ‘wicked’ environmental problems of Oklahoma.
- Analyzed longitudinal panel data from the Meso-Scale Integrated Socio-Geographic Network (M-SISNet) to examine changes in policy beliefs, institutional trust, climate attitudes, and policy learning.
- Conducted focus group discussions and GMB sessions to tackle Oklahoma’s complex environmental challenges, engaging stakeholders in developing causal loop diagrams and mental models to identify sustainable solutions.
- Applied Advocacy Coalition Framework (ACF) and the Narrative Policy Framework (NPF) for analyzing policy processes and environmental governance.
- Assisted in the preparation of research reports, presentations, and journal publications.
- Participated in meetings, workshops, and conferences (research retreats and academies) to share research findings and insights.
- Facilitated the student-peer networking breakout session in the Fall Retreat 2021.

Research Projects and Experience (continued)

Group Model Building – Tribal Nation Community Engagement Project Facilitator & Chart Writer

Fall 2022
Pawnee Nation, Oklahoma

- Facilitated GMB sessions among the key stakeholders of Pawnee Nation Tribal community.
- Collaborated in a dynamic team setting to develop strategies for a ‘Commercial Kitchen’.
- Identified four key drivers for agricultural development: extension, general knowledge, funding, and a business plan, highlighting the overlaps and stressing the importance of each theme.

- Contributed to discussions on sustainability and community support through various initiatives, including regenerative grazing, small-scale vegetable production, and educational outreach.
- Used mental models and causal loop diagrams to understand the themes discussed during the GMB session.
- Identified the drivers and barriers that could promote sustainability in the community.

Environmental Humanities Research Fellow
Center for the Humanities, Oklahoma State University

August 2023 – 2024
Stillwater, Oklahoma

- Developed “Moving Orange in a Green Way”, a project dedicated to promoting environmentally sustainable practices within the Stillwater community.
- Co-developed an upcoming course titled “Environmental Thinking and Philosophy,” aimed at disseminating regional environmental ideologies and policy frameworks.
- Collaborated on the creation of a rideshare application to improve sustainable transit solutions for the campus community.
- Contributed to “Building the Bigger Table”, an initiative designed to encourage dialogue on food security and strengthen community bonds via collective storytelling and shared culinary experiences.

Master’s Dissertation Research
University of East Anglia

December 2018
Norwich, United Kingdom

- Investigated the role of Corporate Social Responsibility (CSR) in Hospitals.
- Performed extensive literature review on the research background and related problems.
- Conducted a qualitative case study on Serco (a public service firm) in Norfolk and Norwich University Hospital.
- Integrated staff questionnaires and analyzed company data and documents.
- Analyzed data using Excel and SPSS.
- Identified various drivers and barriers for the company to engage in environmental projects.

Publications

Peer-Reviewed Publications

1. Sibley, M., Peach, K., León-Corwin, M., **Selvakumar, P. P.**, Diodosio, K., Fox, A., ... & Olofsson, K. (2024). Exploring risk-scapes in Oklahoma: Institutional Trust, Environmental Justice, Climate Change, and Infrastructure. *Safer Communities*.
2. Olofsson, K. L., **Selvakumar, P. P.**, Peach, K., Leon-Corwin, M., Stormer, S. A., Gupta, K., ... & Sibley, M. (2023). Effective Stakeholder Engagement in Environmental Problem-Solving through Group Model Building: An Oklahoma Case Study. *Environmental Challenges*, 100755.
3. **Selvakumar, P. P. (2020)**. Investigation of Corporate Social Responsibility in Multi-National Companies to Promote Environmental Sustainability. *OIDA International Journal of Sustainable Development*, 13(05), 11-20.

Manuscripts Under Review

1. “Characterless: Developing Macro Narratives within the Narrative Policy Framework.” – Submitted to *Policy Studies Journal*.
2. “Home Sweet ’Homa: Place Attachment, Political Participation, and the Environment in Oklahoma” – Manuscript submitted to *Social Science Quarterly*.
3. “From Experience to Expectation: The Normalization of Environmental Harm in an Era of Climate Volatility,” – Manuscript submitted to *Social Problems*.

Manuscripts in Preparation

1. “Beyond the Checklist: Toward Relational Engagement in Coastal Resilience Planning” – Work in progress.
2. “Women at the Intersection of AI and Climate Justice: A Systematic Review” – Work in progress.
3. “Mapping the Literature at the Intersection of AI and Climate Justice” – Work in progress.

4. “What Does Learning Look Like? A Methodological Advancement to Measure Learning in the Advocacy Coalition Framework (ACF)” – Work in progress.
5. “Who Learns How: An ACF Study Exploring Individual Attributes of Climate Change Learners” – Work in progress.

Book Chapters

1. Selvakumar, P., Moore, S., Petriello, M. A., Lee, H., Baird-Zars, B., Gallay, P., & Sanders, V. (under review). *Creating fair and accountable spaces to support community-driven flood risk reduction planning*. Manuscript under review.

Grants, Fellowships, and Research Support

- **“Principal Investigator, Columbia Climate School FY27 Research Seed Fund – \$49,500 (2026)”**. Awarded funding for *A Knowledge, Attitudes, and Practices (KAP)-Informed Flood Risk Communication Pilot in Douala*, supporting the development and implementation of a community-centered flood-risk communication initiative in Douala, Cameroon.
- Earth Network on AI, Climate and Society – Columbia Climate School (2026–Present). **Co-Director**; total project award: \$50,000. Co-lead the Earth Network on AI, Climate and Society, supporting interdisciplinary research, programming, and collaboration at the intersection of artificial intelligence, climate change, sustainability, and society. Director: Dr. Marco Tedesco.
- Robberson Summer Dissertation Fellowship — \$8,000 (2024)
- Received \$1,980 funding from the Fae Rawdon Norris Grant Committee for the "Foods of Home – Building a Bigger Table" event featuring guest speaker Dr. Gisela Heffes for the Fall 2024 – Spring 2025 period.
- Granted the 2023 – 24 Humanities Research Group Fellowship, providing funding of \$3000 to pursue research in the humanities field.
- Received the 'Grand Prize Scientific Poster Presentation' and a \$2000 cash award for delivering a compelling presentation on 'Cross Stakeholder Assessment of Causes and Consequences of Environmental Problems in Oklahoma' at the EPSCoR S3OK Research Retreat.
- Secured a \$2000 travel grant from the Department of Environmental Science to support attendance at conferences and professional development activities.
- Received conference funding of \$300 from the Graduate and Professional Student Government Association to present research findings at a professional conference.
- Received \$200 for second place in Oklahoma State University Abstract Writing Competition.

Teaching Experience

Politics, Policy & Diplomacy of Climate | Columbia University | Fall 2026

Teaching Assistant. Support Ambassador Shefali Razdan Duggal’s interdisciplinary graduate seminar examining climate politics, policy, governance, implementation, diplomacy, and geopolitical challenges. Support discussion-based and applied learning, including student development of evidence-based climate proposals through stakeholder analysis, political feasibility, implementation pathways, and policy defense.

Undergraduate Research Mentorship | Columbia University/Barnard College | 2025–Present

Mentored a senior thesis examining coastal resilience in East Harlem through iterative research development and feedback; continued working with the student to translate the research into a policy brief for community organizations.

AI, Climate and Society | Columbia University | Fall 2026

Course Developer and Facilitator. Co-developed and facilitate course content examining AI applications, environmental impacts, governance, equity, and climate policy for interdisciplinary learners.

Environmental Thinking and Philosophy | Oklahoma State University

Co-developed course content on regional environmental ideologies, sustainability, and policy frameworks through the Environmental Humanities Research Fellowship.

AI Foundation 101 | Columbia Climate School | April 2026

Co-designed interdisciplinary instructional content and facilitated two AI Foundation 101 sessions for approximately 100 participants, translating foundational AI concepts, responsible use, and climate-and-society applications for non-specialist learners.

Selected Summer Schools and Institutes

- **Rutgers University Summer School on Sustainability Governance** | August 2026
Rutgers University, New Brunswick, NJ
- **New Energy Summer Summit** | July 2026
Arthur L. Irving Institute for Energy and Society, Dartmouth College, Hanover, NH
- **Summer School in Computational Social Sciences** | Summer 2024
University of Rochester, Rochester, NY

Conference Presentations

1. **National Sustainability Society Conference 2026** Colorado State University, Denver
'Beyond the Checklist: Toward Relational Engagement in Coastal Resilience Planning'
2. **National Sustainability Society Conference 2025** University of Notre Dame, South Bend, Indiana
'Artificial Intelligence and Climate Justice' - Speed Talk
3. **Midwest Political Science Association 2025** Chicago, Illinois
'Who Learns How: An ACF Study Exploring Individual Attributes of Climate Change Learners'
4. **National Sustainability Society 2024** Seattle, Washington
'Sustainability Performance and Leadership in Higher Educational Institutions: A Comparative Examination between Oklahoma State University (OSU) and Colorado State University (CSU)'
5. **XVIII World Congress of Comparative Education Societies** Cornell University, New York
'Gamifying Campus Sustainability through a Unified Digital Ecosystem'
6. **Oklahoma EPSCoR S30K Annual State Conference 2024** Oklahoma City, Oklahoma
'From Outages to Outlook: Power Outages and Trust in Institutions'
7. **Southern Political Science Association 2023** St. Pete's Beach, Florida
'Changing Your Mind: Testing the Flexibility of Individual Policy Beliefs'
8. **Oklahoma Clean Lakes and Watersheds Association 2023** Stillwater, Oklahoma
'Collaboration for socially sustainable solutions for water issues in Oklahoma.'
9. **Oklahoma EPSCoR S30K Annual State Conference 2022** Tulsa, Oklahoma
'Collaboration for socially sustainable solutions for water, carbon and infrastructure resilience in Oklahoma.'
10. **International Conference on Sustainable Development 2019** Harvard University, Boston
'Investigation of Corporate Social Responsibility in Multinational companies to promote Environmental Sustainability'
11. **OK NSF EPSCoR Research Retreat** Midwest City, Oklahoma
'Cross Stakeholder Assessment of Causes and Consequences of Environmental Problems in Oklahoma'
12. **The Asian Conference on Sustainability, Energy & the Environment 2019** Virtual Presentation
'Role of Corporate Social Responsibility in Hospitals to Improve Environmental Sustainability'

Public Scholarship and Research Communication

- Co-authored public-facing article with Dr. Marco Tedesco, "How Can AI Address Climate Justice When Women's Voices Are Silenced?" published by Columbia Climate School's *State of the Planet*, February 2026. <https://lamont.columbia.edu/news/how-can-ai-address-climate-justice-when-womens-voices-are-silenced>
- Research featured by Dream.Org in "What AI Has To Get Right About Women," highlighting work on women, climate justice, and responsible AI. <https://dream.org/news-articles/ai-has-a-women-problem/>
- Developed perspective essay, "Parenting a Responsible AI for Climate Governance," connecting AI, climate justice, responsibility, and lived experience. <https://news.climate.columbia.edu/2026/06/04/raising-ai-for-a-just-climate-future/>

- Prepared a research report for the Data Science Institute on energy outages, climate vulnerability, socioeconomic risk, and power outage burden in New York City.
- Delivered an invited talk to the MPG/SGT seminar group on research at the intersection of artificial intelligence, climate justice, and community-centered climate resilience planning.

Academic Leadership and Service

Co-Director of Earth Network on AI, Climate and Society, Columbia Climate School 2026 – Present

- Co-lead the Columbia Climate School's Earth Network on AI, Climate and Society, advancing interdisciplinary research and collaboration at the intersection of artificial intelligence, climate change, sustainability, and society.
- Help shape the network's strategic direction, research priorities, and programming while connecting scholars and practitioners across climate science, social science, policy, technology, and related disciplines.
- Develop and coordinate interdisciplinary research initiatives, workshops, seminars, and collaborative activities focused on the opportunities, risks, governance, and societal implications of AI in climate and sustainability contexts.
- Build partnerships across Columbia and with external collaborators to foster research, education, and dialogue on responsible and equitable applications of AI for climate and society.

AI Foundation 101 Workshop Co-Lead April 2026

- Co-developed the AI Foundation 101 workshop with Andrea Paredes, Dr. Marco Tedesco, and Mahdad Parsi for audiences at Lamont-Doherty Earth Observatory and the Columbia Climate School.
- Helped shape workshop goals, session structure, and participant-facing materials to support broader AI literacy across climate, science, and policy communities.
- Expanded program reach by supporting delivery across two sessions and formats, including in-person and online participation.
- Reached approximately 100 participants while strengthening Columbia Climate School programming on responsible AI, climate, and society.

Student Executive Board Member – NSF EPSCoR Spring 2022 – May 2025

- Orchestrated the Student Professional Development Session at the Oklahoma NSF EPSCoR 2024 Annual Conference, facilitating career advancement and professional skills enhancement for attendees.
- Organized a seminar series to foster academic and professional development among peers.
- Organized an OSU/OU Football Game Watch Meet-Up, facilitating networking in an informal setting.
- Contributed to the formulation of the board's mission statement and future strategic initiatives.
- Coordinated the student poster session for the EPSCoR annual meeting in April 2023.

Vice President – Society for Environmental Scientists Fall 2023 – 2025

- Elected as Vice President to steer the club's initiatives, driving engagement and promoting environmental awareness within the university community.
- Led the planning and execution of educational events, workshops, and community service projects.
- Facilitated collaboration between students and faculty to enrich the club's activities and impact.

Professional Experience

Graduate Assistant – Water Quality Testing Summer 2023
City of Stillwater (Water Utilities Department) Stillwater, Oklahoma

- Conducted regular monitoring of water quality parameters in the city of Stillwater.
- Assessed potential contaminants and ensured the safety and compliance of the water supply.

- Implemented a comprehensive water sampling program in coordination with local authorities.
- Collected representative samples from various sources to assess key water quality indicators.
- Identified any potential issues or deviations from established standards.

Project Coordinator – ‘Budget Bites’ Project Sep 2017 – Sep 2018

Enactus UEA

Norwich, United Kingdom

- Conceptualized and implemented ‘Budget Bites’ project, addressing food waste and food poverty in Norwich.
- Sourced surplus food by establishing partnership with the Norwich Food Hub.
- Contributed to the upcycling of collected food items, offering them to the community on a 'pay as you feel' basis to ensure accessibility to affordable food.
- Successfully launched two pop-up stalls, leading to the development of the Budget Bites Café concept aimed at providing affordable meals and work experience opportunities.
- Supported the project's alignment with the United Nations Sustainable Development Goals, specifically targeting responsible consumption and production, zero hunger, and decent work and economic growth.
- Presented the ideas of the project ‘Budget Bites’, an idea of sourcing food surplus and upcycling them for consumption, in Enactus UK National Expo held in London.
- Fostered community relations by organizing events such as pop-up stalls, training, and workshops.

International Students Ambassador

University of East Anglia

Jan 2018 – Nov 2018

Norwich, United Kingdom

- Supported the International Office in administrative activities and responsibilities.
- Assisted International Students with prompt and clear communications about campus developments.
- Networked with prospective students through email accounts and social media platforms.
- Promoted diversity, equity and inclusion by conducting deeper engagement with the future students.
- Arranged campus tours for the incoming students.

Engineer (Sustainable Development Team)

Tata Consultancy Services

June 2015 – Aug 2017

Chennai, India

- Developed user interface to track energy consumption in mobile applications.
- Contributed to rein in company’s carbon and resource footprint using designing and technical skills.
- Used technologies including React.js, JavaScript, Python and CSS.
- Assessed the organization’s sustainability performance using various statistical skills.
- Identified the ‘limited access to energy and water in the campus’ as the organization’s key sustainability issues that led to reduced productivity at the workplace.
- Disseminated awareness about responsible consumption of water and electricity at the workplace through floor walks.
- Collaborated with the company’s internal groups and developed sustainability plans through focus group discussions.

Academic and Professional Service

- Completed eight manuscript reviews across *Social Problems*, *African Research Reports*, and the *International Journal of Sustainability in Higher Education*; Web of Science ResearcherID: IWU-9586-2023.
- Columbia Climate School Research Seminar Series Planning Committee Member, 2026–27 academic year.
- Member, National Society for Sustainability Planning Committee, supporting planning and coordination of national conference activities to be held in Denver, Colorado, 2026.
- Co-led NSS open symposium with Dr. Laura Zanotti, “Storytelling & Brave Spaces for Sustainability: Bridging Institutions and Communities.”
- Provided constructive feedback to undergraduate researchers during the third annual OSU Undergraduate Research Symposium poster session, helping to enhance their presentation and research skills.

- Spearheaded sustainability initiatives as Net Impact Graduate Chair, OSU Chapter, fostering a more inclusive and equitable environment at OSU.

Professional Development

- Participated in Spring 2026 Climate Stories Ambassadors training program, focused on climate storytelling, interviews, and public engagement.
- Attended Data Science and AI Exchange (DAX) 2026, Columbia University, engaging with research and discussions on responsible AI, data science innovation, and interdisciplinary applications of AI.
- Attended the Fall 2024 Graduate Teaching Assistants conference and gained knowledge in effective instructional techniques, contributing to the development of my teaching portfolio.
- Attended conference workshops on Collaborative & International Grants, Leadership in 3D and Improving your Publication Success conducted by KiND Institute.
- Completed a workshop on Closing the loop: why do we need post occupancy evaluation for buildings at the Appalachian State University, Boone, North Carolina (September 2018).
- Participated in the Eastern Arc Start-up Weekender 2019 conducted by Eastern Academic Research Consortium that created an opportunity to drive growth and innovation in the project ideas through various activities and academic expertise.
- Participated in the Barclays Accelerator Programme hosted by key decision makers at Barclays to provide guidance for innovative solutions in technological world.
- Headed the project team that was selected in 'Manthan National Competition' held in 2013. Presented our ideas on 'Can Crusher', a proposal for a mechanism that could reduce crushing cost incurred for cans and bottles.

Awards/Certificates

- Awarded International Student Academic Excellence Scholarship for outstanding academic performance and extracurricular involvement at the University of East Anglia.
- Top Cultural Ambassador Award 2018 of University of East Anglia for demonstrating cultural sensitivity and appreciation through work.
- Top Performer Award 2018 of Enactus UEA for extraordinary contribution for the project 'Budget Bites'
- Special Initiative Award – 2016 of Tata Consultancy Services for spreading awareness through 'floor walks' on responsible consumption of water and electricity at the workplace among the employees.
- Certification in Human Resource and Business Analytics (2015) of Loyola Institute of Business Administration (LIBA), for completing courses in statistical computing, data analytics and scientific research used by researchers to retrieve, clean, analyze, visualize and present data.

Methods, Technical Skills, and Languages

Research Methods & Technical Skills: Proficient in Vensim, Stata, ArcGIS, IBM SPSS, R programming, React.js, Python, JavaScript, CSS, and Microsoft Office.

Language Skills: Native proficiency in Tamil, professional proficiency in English.